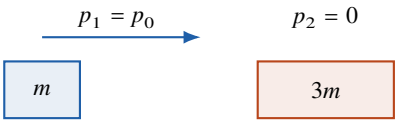
Before contact



$$P = p_0 \qquad A_p = p_0$$
$$K = \tfrac{1}{2}mu^2$$

After elastic release

$$P = p_0 \qquad A_p = 2p_0$$
$$K = \tfrac{1}{2}mu^2$$